INCH-POUND

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MILITARY SPECIFICATION

AIR CLEANERS, INTAKE: DRY-TYPE

(FOR INTERNAL-COMBUSTION ENGINE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers intake air cleaners, including elements of the two-stage dry type for use with internal-combustion engines in stationary applications, industrial and construction equipment, and other specialized applications.

1.2 Classification. The air cleaners shall be of the following classes, as specified (see 6.2):

Class I - 8 hours, 70 percent primary separator efficiency.
Class II - 12 hours, 70 percent primary separator efficiency.
Class III - 15 hours, 90 percent primary separator efficiency (design optional).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 2940

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-B-117	- Bags, Sleeves and Tubing-Interior Packaging.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-43666	- Boxes, Shipping, Consolidation.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking.
MS39247	- Air Cleaner, Intake: Dry Type, for Internal Combustion Engines (Class I and II).

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of

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documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J726 - Air Cleaner Test Code.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Class I and class II air cleaners shall be as shown on MS39247 and as specified herein. Class III air cleaners shall be as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. The material shall be as specified herein. Materials not specified shall be selected by the contractor, and shall be subjected to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The air cleaner shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the air cleaner may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw

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materials. The components, pieces and parts incorporated in the air cleaner may be newly fabricated from recovered materials to the maximum extent practicable, provided the air cleaner produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the air cleaner.

3.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for servicing, repair, and replacement without the use of special tools and without removing or disturbing the clean-air chamber connection to the engine. The number of parts required to be removed for servicing shall be held to a minimum. Parts shall be fabricated so as to prevent incorrect reassembly. Covers or plates which must be removed for adjustment, repair, replacement, or servicing shall be equipped with quick-disconnect fastenings.

3.4.1 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

3.5 Tubular joints. Tubular joints, at the outlet only, shall incorporate an externally beaded end that shall withstand external pressure caused by hose clamps or other fastening devices.

3.6 Maximum rated airflow. Unless otherwise specified (see 6.2), the maximum rated airflow capacity in cubic feet per minute of air shall be as specified in MS39247.

3.7 Performance characteristics. Air cleaners shall have the performance characteristics specified herein.

3.7.1 Resistance to air leakage. The clean-air chamber of an air cleaner shall not leak under 2 psi (pounds per square inch) air pressure or when subjected to a vacuum of 50 ± 1-1/2 inches of water.

3.7.2 Air cleaner restriction. The restriction (see 6.4.1) of a new air cleaner at maximum rated airflow shall be not more than 8 inches of water for class I air cleaners, and 12 inches of water for class II and class III air cleaners.

3.7.3 Air cleaner efficiency. When tested as specified in 4.6.2.5, the air cleaner efficiency shall be not less than 99 percent.

3.7.4 Dust capacity. When tested as specified in 4.6.2.6, the class I air cleaner shall operate for 8 hours, the class II air cleaner for 12 hours, and the class III air cleaner for 15 hours under constant flow with no component servicing. Operation shall be based on 20 inches of water restriction.

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3.7.4.1 Air cleaner efficiency during dust-capacity test. When tested as specified in 4.6.2.6, the accumulative efficiency (see 6.4.2) to 20 inches of water shall be not less than 99.5 percent. The initial and second efficiency shall be not less than 99.5 percent. The accumulative efficiency from 20 to 50 inches of water or 10 hours, whichever occurs first, shall be not less than 99 percent. The specific efficiency (see 6.4.3) at 50 inches of water or 10 hours, whichever occurs first, shall be not less than 99 percent.

3.7.4.2 Air cleaner primary separator efficiency. When tested as specified in 4.6.2.6, the primary separator efficiency shall be not less than 70 percent for classes I and II air cleaners, and not less than 90 percent for class III air cleaners.

3.7.5 Service life. When tested as specified in 4.6.2.7, the service life of class I air cleaner shall be 15 hours with a frequency of element cleaning of not less than 1 hour intervals. The service life of class II and class III air cleaners shall be 25 hours and 30 hours respectively, with a frequency of element cleaning of not less than 1-1/2-hour intervals. The efficiency of any air cleaner shall not be less than 99.5 percent during any cycle. The arithmetic average of the primary separator efficiency shall be not less than 70 percent for classes I and II and 90 percent for class III air cleaners.

3.7.6 Restriction after first cycle of service test. The restriction at maximum rated airflow of the air cleaners after one servicing shall be not more than its original restriction plus 1-1/2 inches of water.

3.7.7 Vibration resistance. When tested as specified in 4.6.2.9, the air cleaners shall not leak or show evidence of structural damage or permanent deformation.

3.7.8 Resistance to water washing. When tested as specified in 4.6.2.10, the filter element shall have a restriction recovery within 1 inch of water of rated value, shall have an efficiency of not less than 99 percent, and shall show no evidence of damage or permanent deformation.

3.7.9 Fungus resistance. The air-cleaner element shall not be adversely affected when tested as specified in 4.6.2.11. On completion of the test, the element shall conform to 3.7.2 and 3.7.3.

3.8 Identification marking. The air cleaners shall be identified in accordance with MIL-STD-130, and shall include the military part number (see 6.5).

3.9 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the air cleaner and its components and parts normally painted shall be cleaned, treated, and painted, in accordance with MIL-T-704, type A, color optional.

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3.10 Workmanship. The air cleaner shall be free of burrs, sharp edges, surfaces out of alignment or out of contour (visually apparent), paint sags, runs, blisters, or incomplete coverage, and loose metal particles and other foreign matter.

3.10.1 Sheet metal parts. Sheet metal parts shall be free of pits, laminations, blisters, tears, visual metal thinning at drawn sections, and sharp edges.

3.10.2 Filtering element. The filtering element shall be free of loose material and shall be firmly positioned in its housing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Inspection of packaging (see 4.7).

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4.3 First article inspection. Unless otherwise specified (see 6.2), two first article air cleaners shall be examined as specified in 4.6.1 and subjected to the tests marked "X" in column 1 of table I in the order listed. Presence of one or more defects or failure of any test shall be cause for rejection.

TABLE I. Test schedule.

First Article	Quality Conformance	Comparison	Test	Test paragraph
1	2	3	4	5
X	X	X	Resistance to air leakage.	4.6.2.3
X	X	X	Airflow restriction.	4.6.2.4
X	-	X	Efficiency at max. rated airflow.	4.6.2.5
X	-	X	Dust capacity at constant airflow.	4.6.2.6
X	-	-	Service life.	4.6.2.7
X	-	-	Restriction after first cycle of service test.	4.6.2.8
X	-	X	Vibration.	4.6.2.9
X	-	-	Water washing.	4.6.2.10
X	-	-	Fungus.	4.6.2.11

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.2 Examination. Sample selected in accordance with 4.4.1 shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection. An additional element shall be used for the dust capacity test at constant airflow (see 4.6.2.6).

4.5 Inspection comparison. The Government may select air cleaners at any time during the contract production period and subject these air cleaners to the examination specified in 4.6.1 and tests marked "X" in column 3 of table I in the order listed. The inspection will be performed by the Government, at a site selected by the Government, on air cleaners selected at random from those which have been accepted by the Government and will not include the previously inspected first article air cleaners. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection

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comparison and failure of such additional tests shall have the same effects as failure of those tests specified as inspection comparison. An additional element shall be used for the dust capacity test (see 4.6.2.6).

4.5.1 Inspection failure. Failure of an inspection comparison air cleaner to meet any requirement specified herein during and as a result of the examination specified in 4.6.1 and tests marked "X" in column 3 of table I, shall be cause for rejection of the inspection comparison air cleaner and shall be cause for refusal by the Government to continue acceptance of production air cleaners until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on air cleaners previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all air cleaners accepted prior to the completion of the inspection comparison are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Inspection procedure.

4.6.1 Examination. The air cleaner shall be examined as specified herein for the following defects:

101. Dimensions not as specified (see MS39247).
102. Materials not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable operating and storage environment (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
105. Contractor does not have documentation available for identification of materials, material finishes, or treatments (see 3.3.1.2).
106. Used, rebuilt or remanufactured components, pieces, and parts incorporated in the air cleaner (see 3.3.2).
107. Maintenance provisions not as specified (see 3.4).
108. Major assemblies and attachments not interchangeable (see 3.4.1).
109. Bent or otherwise defective tubular and flanged joints (see 3.5).
110. Maximum airflow not as specified (see MS39247 and 3.6).
111. Identification marking not as specified (see 3.8).
112. Treatment and painting not as specified (see 3.9).
113. Color not as specified (see 3.9).
114. Workmanship not as specified (see 3.10).
115. Filtering element not firmly positioned in its housing (see 3.10.2).

4.6.2 Tests.

4.6.2.1 Test conditions.

4.6.2.1.1 Temperature. The ambient temperature during tests shall be 75 °F ± 15 °F.

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4.6.2.1.2 Relative humidity. The relative humidity during tests shall be 50 percent ± 15 percent.

4.6.2.1.3 Airflow. Specified values of airflow are for the standard atmospheric conditions of 29.92 inches Hg barometric pressure and 80 °F ambient temperature. Actual airflow values shall be corrected to standard conditions and shall be within ± 2 percent of specified values.

4.6.2.1.3.1 Pulsation or aspirator airflow. When the dust unloader or aspirator connection type air cleaners are tested, a 10 percent of rated cubic feet per minute (cfm) air supply shall be provided for the pulsation or aspirator flow test condition.

4.6.2.1.4 Test dust. The chemical analysis and the particle size distribution by weight of the test dust shall be as specified in SAE J726.

4.6.2.1.4.1 Dust feeding rate. When the feeding of dust is required as part of any test, the dust shall be fed at an average rate of 0.025 ± 0.005 gram per cubic foot of air, unless otherwise specified herein.

4.6.2.1.5 Air cleaner preparation. Unless otherwise specified herein, a clean unused filter element shall be installed in the air cleaner for each test (the filter element is not considered "used" after use in a restriction test). Further, the air-cleaner assembly (less filter element) shall be cleaned thoroughly with compressed air if it has been subjected previously to a test in which dust was used.

4.6.2.1.6 Air cleaner mounting position. Unless otherwise specified herein, the air cleaner shall be mounted in a vertical position for all tests specified herein.

4.6.2.2 Test equipment.

4.6.2.2.1 Piezometer, adapter, manometer, and flowmeter. Piezometer conforming to SAE J726 and equal to or the next size larger (maximum of 1/2 inch larger) than the air cleaner inlet shall be installed at the inlet and outlet of the air cleaner. Transition ducts (if required) shall be installed from the flanges of the air cleaner inlet and outlet to the piezometers. Adapters installed after the outlet piezometers shall have sides at angles not more than 7 degrees as measured from the centerline of the transition piece. A smooth approach (ideal flow orifice specified in SAE J726) shall be installed on the front of the inlet piezometer tube. A manometer (water type) shall be installed between the two piezometers as shown in figure 2A of SAE J726. A flowmeter shall be connected to the outlet of the adapter.

4.6.2.2.2 Master filter. The master filter shall consist of one or more layers of 1/2-inch fiberglass batting (Manville Corp. AFS-4 backed media, or equal) installed in a housing with the nap side of the fiberglass batting facing upstream. The master filter batting shall be of a size to prevent restriction (pressure drop) to the airflow before the test is completed. A new master filter

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batting shall be installed if, during any service life or dust capacity test, the air-pressure drop across the master filter increases 10 inches of water above its original restriction. (The face velocity of air against the master filter batting shall be not more than 165 feet per minute. A manometer shall be located in the test setup to measure the pressure drop across the master filter in order to determine when the static pressure across the master filter increases 10 inches of water above its normal restriction). SAE J726 shows the test setup for dust feeding and the dust feeder. The filter housing shall be grounded during tests.

4.6.2.2.3 Preparation of master filter. The fiberglass batting shall be hung in an air oven and dried for 6 hours at a temperature of 200 °F to 240 °F. This drying time may be reduced and the batting considered as dry if there is no change in weight of the batting at two consecutive 10-minute intervals. A similar drying and weighing procedure shall be conducted after each test.

4.6.2.3 Resistance to air leakage. One of the following air leakage tests shall be used to determine resistance to air leakage at the contractor's option.

4.6.2.3.1 Air leakage test number 1. Seal the inlet to prevent air leakage at this location. Connect the outlet of the air cleaner to a flowmeter with a maximum flow reading not more than 5 cubic feet of air per minute. Accuracy of flow shall be plus or minus 1/10 of 1 percent of the air cleaner rated flow. Assure tight flowmeter connections. Subject the air cleaner to a vacuum of 50 plus or minus 1-1/2 inches of water as measured at the air cleaner outlet. Evidence of any airflow shall constitute failure of this test.

4.6.2.3.2 Air leakage test number 2. Seal the inlet and the outlet of the clean-air chamber and pressurize the interior of the clean-air chamber to 2 pounds per square inch air pressure through a pipe coupling connected to the drilled and tapped hole in the inlet or outlet sealing plate. Submerge the air cleaner in water. Any evidence of air bubbles escaping from the air cleaner shall constitute failure of this test.

4.6.2.4 Airflow restriction. Conduct this test on a test setup specified in SAE J726. Set the airflow at maximum rated flow and record the restriction. Nonconformance to 3.7.2 shall constitute failure of this test.

4.6.2.5 Efficiency at maximum rated airflow. Prepare the air cleaner as specified in 4.6.2.1.5. Install a master filter as specified in 4.6.2.2.2, start the airflow and adjust to maximum rated flow. If one master filter does not contain sufficient batting area to meet the requirements of 4.6.2.2.2 (165 feet per minute), two or more shall be installed in parallel, as applicable. Feed fine grade test dust as specified in 4.6.2.1.4 into the air cleaner inlet at the rate of 0.025 gram of dust per cubic foot of air until 30 minutes of feeding shall have elapsed. Remove the master filter taking care to retain all entrapped dust. Obtain the efficiency by means of the following formula:

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$$\text{Efficiency, percent} = \frac{W_1 - (W_2 - W_3)}{W_1} \times 100$$

Where W_1 = Weight of dust fed into air cleaner inlet.
W_2 = Weight of master filter and outlet conductors (see 6.4.4).
W_3 = Original weight of master filter and outlet conductor.

Note. All dust measurements shall be accurate to the nearest 0.01 gram. Other measurements shall be accurate to nearest 0.01, wherever possible, but never less than to the nearest 0.1 gram. Nonconformance to 3.7.3 shall constitute failure of this test.

4.6.2.6 Dust capacity at constant airflow. After pretest preparation (see 4.6.2.1.5), install a master filter (see 4.6.2.2.2) between the air cleaner and the flowmeter. Weigh the filter element (W_5). Start airflow and adjust to the maximum rated flow of the air cleaner. Measure and record the restriction at start and at 10-minute intervals throughout the test. Feed coarse grade dust, as specified in 4.6.2.1.4, into the air inlet for 30 minutes at the rate of 0.025 gram of dust per cubic foot of air. Weigh the master filter in accordance with 4.6.2.2.3. At the end of the first 30 minutes, record the initial specific efficiency. After completing the initial specific efficiency test, replace the fiberglass batting and continue feeding dust to the air cleaner to a 20 inches of water restriction and then weigh and replace the fiberglass batting and the filter element. Determine primary separator efficiency by means of the following formula:

$$\text{Primary separator efficiency, percent} = \frac{W_1 - (W_2 - W_3) - (W_4 - W_5)}{W_1} \times 100$$

Where W_1, W_2, W_3, and accuracy of dust measurements are as defined in 4.6.2.5.

W_4 = Weight of filter element.
W_5 = Original weight of filter element

Conduct a second 30-minute specific efficiency test. After completing this second specific efficiency test, replace the fiberglass batting and continue feeding dust to the air cleaner to a 50 inches of water restriction (or 10 hours, whichever occurs first) and weigh and replace the fiberglass batting. Conduct a third and last specific efficiency test for 30 minutes or to a maximum of 70 inches of water, whichever occurs first. Nonconformance to 3.7.4, 3.7.4.1, or 3.7.4.2 shall constitute failure of this test.

4.6.2.7 Service life. Following completion of the tests specified in 4.6.2.6, empty the air cleaner dust cup, if used, and replace the filter element with a new element. Weigh the new filter element (W_5). Reassemble the cleaner and determine and record the initial restriction. Conduct a constant maximum rated airflow dust capacity test at a dust concentration of 0.100 gram per cubic foot of airflow, to a total restriction of 20 inches of water. Then clean the air

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cleaner in accordance with 4.6.2.7.1. Repeat the procedure until a total test time of 15 hours for class I, 25 hours for class II, and 30 hours for class III air cleaners are accumulated, or until the frequency of cleanings exceeds that specified in 3.7.5, whichever occurs first. Clean the air cleaner dust cup, if used. Weigh the filter element (W_4) and clean. Determine initial restriction prior to initiation of each dust capacity test. Record the weight of the dust fed for each test. A fiberglass batting shall be installed in the master filter frame at all times. Replace and weigh at the end of each dust capacity cycle, or when the net static pressure rise across the master filter reaches 10 inches of water. Clean the air cleaner element after each cycle in accordance with 4.6.2.7.1. Determine the primary separator efficiency for each test cycle. The arithmetic average of all separator efficiencies shall constitute the total test separator efficiency. Nonconformance to 3.7.5 shall constitute failure of this test.

4.6.2.7.1 Cleaning procedure. Clean the filter element as follows: Use compressed air of approximately 90 psi and blow a stream of dry clean air against the downstream side of the element making certain to avoid damaging filter media. Move the nozzle up and down blowing air through the media and slowly rotate element. Continue cleaning for not less than 20 minutes for rated airflows up to 300 cfm and for 30 minutes for rated airflow above 300 cfm. Cleaning may be stopped prior to the times specified herein if no further dust is being removed from the element as determined visibly. After the filter element is clean, reinstall in the air cleaner.

4.6.2.8 Restriction after first cycle of service test. After the first cycle of the service life test has been completed, remove the filtering element and clean in accordance with 4.6.2.7.1. After the filter element is cleaned, reinstall in the air cleaner. Start the airflow and adjust to the maximum rated airflow (cfm). Determine and record the restriction. Nonconformance to 3.7.6 shall constitute failure of this test.

4.6.2.9 Vibration. Mount the air cleaner on a vibrating machine. Subject the air cleaner to a simple harmonic motion having an amplitude of 1/32 inch (1/16 inch total excursion). Uniformly vary the frequency between the limits of 10 and 40 cycles per second. The entire frequency range from 10 to 40 cps and return to 10 cps shall be traversed in 1 minute. Vibrate the air cleaner for 12 hours in a vertical axis followed by 12 hours in a horizontal axis except that 6 hours shall be in one horizontal axis and 6 hours in the other horizontal axis. Then repeat the resistance to air leakage test specified in 4.6.2.3. Nonconformance to 3.7.7 shall constitute failure of this test.

4.6.2.10 Water washing. Air clean (see 4.6.2.7.1) the element used in the dust capacity test (see 4.6.2.6) and immerse it in water for 1 minute. Agitate for 1 minute in the water. Remove and flush with cool water for 1 minute in a direction opposite to element airflow so that the pressure of the water does not damage the element. Allow the water to drain from the element for 5 minutes. Install the element in the air cleaner and operate at 150 percent of maximum rated airflow for 5 minutes or the element may be oven dried. Repeat the water washing and drying procedure eight times. At the end of the last drying period,

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record the air cleaner restriction at maximum rated airflow. Continue maximum rated flow for 1 hour or until the restriction has recovered to within 1 inch of water of its rated value based on initial restriction from the dust capacity test. Then conduct the efficiency test specified in 4.6.2.5. Examine the filter element. Nonconformance to 3.7.8 shall constitute failure of this test.

4.6.2.11 Fungus. Subject the air cleaner element to method 508, MIL-STD-810. Nonconformance to 3.7.9 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

116. Containers not as specified for level A or level B (see 5.1.1, 5.2.1 and 5.2.2).
117. Box closure for preservation and packing not as specified for level A or level B (see 5.1.1, 5.2.1 and 5.2.2).
118. Air cleaners of unlike description packed together for level A or B.
119. Quantities packed together exceed the weight or size limitation specified for level A or B.
120. Preservation and packing not in accordance with the referenced document for level C.
121. Marking missing, illegible, incorrect or incomplete for level A, B, or level C.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or level C as specified (see 6.2).

5.1.1 Level A. Each air cleaner shall be sealed in a waterproof bag conforming to MIL-B-117, class B and then placed in a close fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Level C. Air cleaners shall be preserved in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, level B or level C as specified (see 6.2).

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5.2.1 Level A. Air cleaners of like description, preserved as specified in 5.1.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional or conforming to PPP-B-621, class 2, in quantities not to exceed the size or weight limitations of the box. Box closure and strapping shall be as specified by the appendix to the box specification except the strapping shall be zinc-coated.

5.2.2 Level B. Air cleaners of like description, preserved as specified in 5.1.1, shall be packed together in a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant, grade optional, or PPP-B-640, class 2 or MIL-B-43666. The gross weight or size shall not exceed the limitations of the box, in accordance with the box specification. Closure, waterproofing, sealing and reinforcing shall be in accordance with the appendix to the applicable container specification.

5.2.3 Level C. Air cleaners, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking specified by the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful,but is not mandatory.)

6.1 Intended use. The air cleaners are intended for use with internal combustion engines for automotive and industrial applications. The application recommendations shown in table II are to be used as a guide in the selection of air cleaners for end items.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required for inspection and approval, and the number of units required (see 3.2, 4.3 and 6.3).
e. Maximum rated airflow required when other than as specified (see 3.6).
f. Treatment and painting if other than as specified (see 3.9).
g. Levels of preservation and packing required (see 5.1 and 5.2).
h. Any special marking required (see 5.3).
i. Specification part or identifying number required (see 6.5).

6.3 First article. When a first article inspection is required, the air cleaner should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the

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TABLE II. Recommended end-item selections.

Class I (8 Hours, 70% Primary Efficiency)	Class II (12 Hours, 70% Primary Efficiency)	Class III (15 Hours, 90% Primary Efficiency)
Batching Plants. 1/	Augers, Earth.	Asphalt plants.
Compressors, Spray-Type, 5 to 25 cfm.	Bins, Aggregate Loading.	Compressors, Reciprocating & Rotary, Rock Drilling & Quarying, 365 to 600 cfm/100 psi.
Curing Machines.	Compressors, Construction Type, 125 to 250 cfm/100 psi.	Cranes & Rough-Terrain & Shovel.
Distributors, Tank-Type.	Cranes, Wheel-Mounted	Crushers & Crushing Plants.
Drier-Mixers, Concrete Mtls.	Crane, Shovel.	Driers, Aggregate.
Finishing Machines, Concrete Paving.	Dehydrators, Sand Compression.	Loaders (Large).
Generator Sets	Ditching Machines.	Rock Drillers, Self-Contained.
Graders, Road. 1/	Dust-Collecting Machines, 2/ Earth-Boring, Rotary.	Scrapers (Large and Rear Engine).
Kettle, Heating, Bitumen.	Excavators, Multipurpose.	Scrubbers & Washers, Aggregate.
Mixers. 1/	Fork-Lift Trucks.	Tractors, Wheeled, Industrial.
Pavers. 1/	Loaders.	Tractors, Full-Tracked, Low-Speed.
Pumps, Concrete.	Scrapers, Small.	
Rollers, Motorized.	Tractors, Crawler (less than 100 H.P.).	
Saw. 1/	Tractors, Wheeled, Agricultural.	
Spreaders.	Trucks, Off-Highway.	
Tractors, Farm & Warehouse.		
Trucks, Over-Highway.		
Welders.		
Winches, Drum.		

1/ Class I or Class II, as applicable.

2/ Class II or Class III, as applicable.

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first article test results and disposition of the first articles. Invitation forbids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Restriction. The static pressure drop across the air cleaner at the observed airflow, expressed in inches of water.

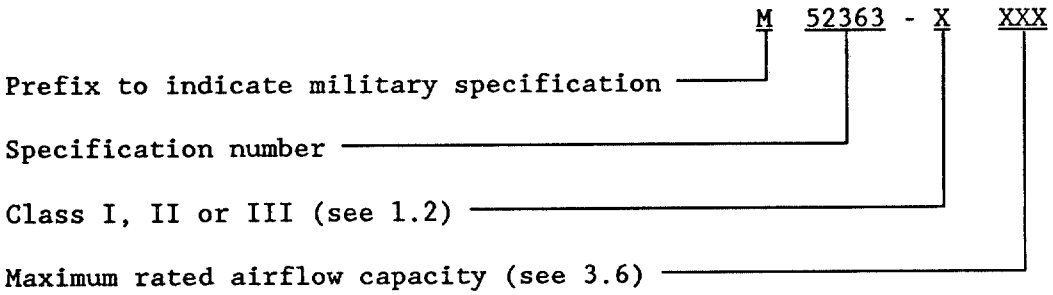
6.4.2 Accumulative efficiency. The accumulative efficiency is based on the total amount of dust fed and includes one or more specific efficiency tests plus the interval between.

6.4.2.1 Accumulative efficiency basis. The accumulative efficiency to 20 inches water restriction and from 20 to 50 inches water restriction shall be based on the total dust fed in the respective test periods.

6.4.3 Specific efficiency. The efficiency between any specified interval, the interval being defined as either time or restriction rise.

6.4.4 Outlet conductors. Outlet conductors are everything downstream from the air cleaner to the master filter batting.

6.5 Part or identifying number (PIN). The PIN to be used for air cleaners acquired to this specification are created as follows:



Example: M52363-I135 is an air cleaner per MIL-A-52363, class I, 135 cfm maximum rating.

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6.6 Subject term (key word) listing.

Air filter
Dust remover
Filter
Precleaner

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - CS

Project 2940-0140

User activities:
Army - AT
Navy - MC, AS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

he preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should oe given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-52363C	2. DOCUMENT DATE (YYMMDD) 910923

3. DOCUMENT TITLE Air Cleaners, Intake: Dry-Type (For Internal-Combustion Engine)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1806 (2) AUTOVON 354-1806
c. ADDRESS (Include Zip Code Army Belvoir RDE Center ATN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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